

Reock Score in Algorithmic Redistricting: Minimum Enclosing Circles, Implementation Proxies, and Moving-Knife Compactness Search

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Abstract

The Reock score compares a district’s area to a circle enclosing the district, providing a compactness measure that is substantially less sensitive to boundary representation artifacts than the Polsby-Popper score. Canonical Reock uses the true minimum enclosing circle; the current BISECT compactness report intentionally uses the centroid-plus-maximum-boundary-radius approximation inherited from the archived Python pipeline, while the moving-knife structure search uses a separate projected-centroid point-cloud proxy. We study reported Reock values across four BISECT structure algorithms—standard-bisect, prime-factor, ratio-optimal, and moving-knife—for North Carolina ($k = 14$), Wisconsin ($k = 8$), and Texas ($k = 38$) under 2020 Census data. Key findings: (1) The reported Reock proxy decreases by ≤ 0.02 when switching from smoothed to raw TIGER/Line boundaries on NC coastal districts, versus PP decreasing by 0.04–0.08, supporting the proxy’s boundary-insensitivity claim. (2) The moving-knife algorithm (MKA), an implemented directional sweep that scores candidate prefixes by minimum point-cloud Reock proxy, achieves mean reported Reock $\geq 0.38^\dagger$ on NC 2020 versus $0.31^\dagger$ for standard-bisect—a 23% empirical improvement in mean Reock. (3) The empirical reported-Reock-PP gap averages +0.15 on NC 2020 under ratio-optimal, showing metric disagreement in this fixture rather than a universal Reock-over-PP ordering. All reported district Reock values use the centroid approximation; gaps may differ from true MBC-based Reock and should be treated as implementation-parity evidence, not exact-MBC evidence. All results use single seed = 42[†].

Keywords

Reock Score, Compactness, Minimum Enclosing Circle, Centroid Approximation, Moving-Knife Algorithm, Gerrymandering, Redistricting

1 Introduction

The Reock score was introduced by Reock [1961] as a measure of how much of an enclosing circle a district fills. Its key conceptual insight is that the minimum bounding circle (MBC) is determined entirely by the most extreme points of a district—its “tips”—rather than by the full boundary perimeter. This makes Reock insensitive to boundary digitisation artifacts, coastline irregularities, and river boundaries that strongly affect perimeter-based metrics such as Polsby-Popper.

1.1 Reock in Court

Reock has been cited in redistricting litigation in several states:

- **North Carolina:** Expert reports in *Harper v. Hall* (2022) cited Reock scores alongside PP values. The NC state supreme court’s compactness analysis referenced the general principle that compact districts should “fill” their bounding geometry [North Carolina Supreme Court, 2022].
- **Wisconsin:** Expert reports in *Gill v. Whitford* (2018) included Reock computations for Assembly districts under the challenged legislative plan [U.S. Supreme Court, 2018].
- **Georgia:** Redistricting litigation in Georgia has produced expert reports using Reock for House and Senate districts.
- **Florida:** The Florida Supreme Court-ordered redistricting (2015–2016) included Reock-based analysis by court-appointed experts.

1.2 Why Reock for Coastal States

For states with Atlantic, Gulf, or Pacific coastlines, PP is unreliable because coastal boundaries contribute zigzag perimeter without corresponding district “mass.” Reock avoids this problem: the MBC is determined by the extreme points (typically inland or cross-bay points), and the jagged coastline between those extreme points does not affect the MBC. This paper

compares the current BISECT Reock proxy’s boundary sensitivity against PP boundary sensitivity under controlled computational conditions. It keeps the canonical MBC definition visible while separating it from the deployed centroid approximation.

1.3 Moving-Knife Algorithm

The moving-knife algorithm (MKA) is a BISECT structure algorithm that uses projected tract centroids to search for directional bisection cuts with strong worst-side compactness. It is not a global Reock optimizer and it does not use polygon exact-MBC Reock inside the splitter. The implementation projects centroids, sorts units by candidate direction, cuts a population prefix, scores both sides with a point-cloud minimum-enclosing-circle proxy, and keeps the best worst-side candidate before local rebalance.

2 Mathematical Definition

2.1 Formula

Let D be a district polygon and let $\text{MBC}(D)$ be the smallest circle that contains D , with radius $r(D)$. The canonical Reock score is:

$$\text{Reock}(D) = \frac{A(D)}{\pi r(D)^2}, \quad (1)$$

where $A(D)$ is the area of D .

Proposition 1. *Reock(D) $\in (0, 1]$ for any non-degenerate polygon D , with equality if and only if D is a disk.*

Proof. By definition, $D \subseteq \text{MBC}(D)$, so $A(D) \leq \pi r(D)^2$. Thus $\text{Reock}(D) \leq 1$. Equality holds iff $D = \text{MBC}(D)$ (as a set), i.e., iff D is a disk. $\text{Reock}(D) > 0$ for any non-degenerate polygon since $A(D) > 0$. □ □

2.2 Canonical Values

2.3 Properties

Proposition 2. *Reock is scale-invariant, rotation-invariant, and translation-invariant.*

Proof. Scaling by λ : $A \mapsto \lambda^2 A$ and $r \mapsto \lambda r$, so $\text{Reock} = \lambda^2 A / (\pi \lambda^2 r^2) = A / (\pi r^2)$. Rotation and translation do not change areas or distances. □ □

Table 1: Reock score for canonical shapes.

Shape	Reock	Approximate value
Disk (circle)	1	1.000
Square of side a	$2/\pi$	0.637
Regular hexagon	$3\sqrt{3}/(2\pi)$	0.827
1×3 rectangle	$4/(5\pi)$	0.255
Equilateral triangle	$3\sqrt{3}/(4\pi)$	0.413

For highly elongated convex polygons, PP tends to exceed Reock; for near-circular shapes, Reock tends to exceed PP. (For example, a 1×6 rectangle has Reock ≈ 0.21 and PP ≈ 0.38 , while a square has Reock ≈ 0.64 and PP ≈ 0.785 .) No uniform ordering between Reock and PP holds for all convex polygons.

2.4 Implementation Boundary

The formula above is the mathematical reference point. The production BISECT compactness report currently computes

$$\widehat{\text{Reock}}(D) = \frac{A(D)}{\pi \hat{r}(D)^2},$$

where $\hat{r}(D)$ is the maximum distance from the polygon centroid to an exterior boundary coordinate. This preserves parity with the archived Python pipeline and is documented in `crates/bisect-analysis/src/compactness.rs`. It is a containing-circle proxy, not the true MBC for every polygon. A separate helper in `crates/bisect-cli/src/bisection_runner.rs` computes a minimum enclosing circle over tract-centroid point clouds for moving-knife candidate scoring; that helper is not the district-polygon compactness metric reported by `bisect label-analyze`.

3 Minimum Enclosing Circle and BISECT Implementation Paths

Computing the exact minimum bounding circle (MBC) of a polygon can be reduced to computing the smallest enclosing disk (SED) of the relevant boundary point set, solvable in expected linear time by Welzl’s randomised algorithm [Welzl, 1991]. That algorithm is the correct reference for exact-MBC Reock, but it is not the same as every Reock-like score exposed by BISECT.

3.1 Implementation paths in bisect

Table 2: Reock-related implementation paths in BISECT.

Surface	What it computes	Evidence boundary
<code>bisect-analysis::reock</code>	Polygon area divided by a centroid-plus-max-boundary-radius circle.	Production compactness metric; Python-parity proxy, not exact MBC.
<code>bisection_runner::welz</code>	Minimum enclosing circle of a projected tract-centroid point set.	Moving-knife candidate-scoring helper; not district-polygon Reock.
Canonical K.2 definition	Polygon area divided by true minimum enclosing circle area.	Mathematical target and external-software comparison point.

The production implementation in `crates/bisect-analysis/src/compactness.rs` uses a **centroid-plus-max-radius approximation**: the circle centre is taken as the polygon centroid, and the radius is the maximum distance from the centroid to any exterior boundary coordinate. This approximation matches the convention used in the Python redistricting pipeline that BISECT supersedes, ensuring comparability of Reock scores across pipeline versions.

Remark 3 (Future work). *Exact polygon MBC remains future work for the compactness report. The centroid-plus-max-radius approximation yields a circle that contains the polygon but may be larger than the true MBC (the approximation errs toward over-coverage, producing Reock values that are conservative—i.e., slightly lower than true-MBC Reock). Practitioners requiring exact MBC-based Reock (e.g., for reproduction against other redistricting software) should note this approximation and should not cite `bisect label-analyze` Reock values as exact-MBC scores.*

Test invariants (L0):

- Reock of a circle is ≤ 1.0 (the approximation may give $\text{Reock} < 1$ for a circle if the centroid is not the true MBC centre; for a centred circle the approximation is exact).
- Approximation circle contains all input boundary points: for all p on the boundary, $\|p - \text{centroid}\| \leq r + 10^{-9}$.
- Approximation circle area $\geq$ polygon area for any convex polygon.

4 Boundary Insensitivity Analysis

We quantify the reported Reock proxy’s boundary insensitivity by comparing values computed on smoothed vs. raw TIGER/Line district geometries.

4.1 Boundary Representations

Raw TIGER/Line. The 2020 Census TIGER/Line tract polygons represent coastlines, rivers, and state boundaries with the full digitisation detail of the source data. NC coastal tracts may have $> 1,000$ vertices along a single coastal segment.

Smoothed boundaries. We apply Douglas-Peucker simplification [Douglas and Peucker, 1973] with tolerance $\epsilon = 100$ metres to reduce vertex counts by approximately 60–80% for coastal tracts, while preserving the overall shape of inland boundaries. This represents a simplified-boundary comparison, not a court-certified “true” compactness value.

4.2 Reock vs. PP Sensitivity

Table 3 compares the change in mean district reported Reock and PP when switching from smoothed to raw boundaries for NC coastal districts (seed = 42[†], ratio-optimal).

Table 3: Mean metric change (smoothed $\rightarrow$ raw boundaries) for NC coastal districts under ratio-optimal, seed = 42[†].

Metric	Smoothed	Raw (TIGER/Line)	Change
Reported Reock	0.374 [†]	0.355 [†]	−0.019
PP	0.261 [†]	0.213 [†]	−0.048

Note: “Coastal districts” = NC districts with $\geq 10\%$ of their boundary on Atlantic coast or sound (approximately 5 of 14 NC districts).

[†]Single-run result, seed = 42.

The reported Reock decrease (−0.019) is $2.5\times$ smaller than the PP decrease (−0.048). This supports the operational claim that the current BISECT Reock proxy is less sensitive to boundary representation than PP for these coastal and riverine districts. It does not by itself prove an exact-MBC Reock sensitivity bound.

4.3 Mechanism

For exact-MBC Reock, the circle is determined by a small support set of boundary points, often two or three extremes. For the current centroid proxy, the radius is determined by

whichever boundary coordinate is farthest from the polygon centroid. In both cases, small coastline wiggles usually matter less than they do for a perimeter sum, but the implemented proxy should be described by its centroid-radius rule rather than by Welzl support-point mechanics.

In contrast, PP depends on the total perimeter $P = \sum_i |e_i|$ summed over all boundary edges. Each additional vertex introduced by TIGER/Line digitisation of the coastline adds perimeter length, directly increasing P and decreasing PP.

4.4 Exact-MBC Smoke Evidence

The repository now includes a separate exact polygon-MBC reference path and a hash-bound smoke package at `docs/examples/k-reock-evidence-packages/exact-mbc-smoke/`. The package contains simple polygon fixtures with known minimum bounding circles: a unit square where exact-MBC and the centroid proxy coincide, and a 3-4-5 right triangle where the exact-MBC circle is supported by the hypotenuse endpoints and the centroid proxy is lower. The verifier lives in `crates/bisect-analysis/src/compactness_evidence.rs`.

This package upgrades the exact-MBC reference evidence, but it does not change the production output of `bisect label-analyze`. District-scale exact-MBC claims still require packages built from real projected district polygons.

5 Moving-Knife Algorithm and Reock Maximisation

The moving-knife algorithm (MKA) is a structure algorithm in `BISECT (--structure moving-knife)` designed to choose directional recursive splits with a strong worst-side compactness proxy.

5.1 Algorithm Description

MKA adapts the moving-knife intuition [Brams and Taylor, 2000] to a recursive tract graph splitter. At each recursive step, the implementation tests a grid of candidate directions, projects tract centroids onto each direction, sorts units by projection rank, and cuts the first population-balanced prefix. It then scores the two sides with a point-cloud enclosing-circle proxy and selects the direction with the best worst-side score before bounded local rebalance.

Formally, at step t of the recursion with region G_t to be divided into k_1 and k_2 districts:

1. Generate candidate directions $\{\theta_1, \dots, \theta_m\}$.
2. For each direction, project centroids, sort units, and form a population-balanced prefix/suffix cut.

3. Compute a proxy score $\text{ReockProxy}^{\min}(\theta_j) = \min(\widehat{R}(L_j), \widehat{R}(R_j))$ over tract-centroid point clouds.
4. Select $\theta^* = \arg \max_j \text{ReockProxy}^{\min}(\theta_j)$, then run the local rebalance path.

5.2 Reock Maximisation Property

MKA does not guarantee a globally maximum Reock partition and does not optimise the exact polygon-MBC metric. It provides an auditable directional baseline whose candidate ledger can be checked against the implemented point-cloud proxy.

5.3 Empirical Validation

Table 4 reports mean and minimum district Reock-proxy values for all four algorithms on NC 2020 (seed = 42[†]).

Table 4: District reported Reock under four structure algorithms, NC 2020, seed = 42[†].

Algorithm	Min	Median	Mean	Max
standard-bisect	0.21	0.29	0.31 [†]	0.46
prime-factor	0.24	0.32	0.34 [†]	0.49
ratio-optimal	0.26	0.35	0.37 [†]	0.51
moving-knife	0.29	0.37	0.38 [†]	0.53

[†]Single-run result, seed = 42.

MKA achieves mean reported Reock = 0.38[†] on NC 2020—a 23% improvement over standard-bisect (0.31[†]) and a 3% improvement over ratio-optimal (0.37[†]). The minimum district Reock under MKA is 0.29, versus 0.21 for standard-bisect (a 38% improvement in the worst-case district). This supports the weaker implementation claim: MKA’s directional sweep is aligned with its worst-side compactness proxy on this single-run fixture. Stronger same-metric or statewide superiority claims require archived per-node candidate ledgers and exact polygon-Reock comparisons.

6 Empirical Comparison Across Structure Algorithms

6.1 Key Observations

WI consistently outperforms NC and TX. Wisconsin’s compact state shape produces higher reported Reock values across all algorithms. The WI moving-knife mean reported

Table 5: District-level reported Reock-proxy statistics by structure algorithm and state (2020 Census, seed = 42[†]).

State	Algorithm	Min	Median	Mean	Max
NC ($k = 14$)	standard-bisect	0.21	0.29	0.31 [†]	0.46
	prime-factor	0.24	0.32	0.34 [†]	0.49
	ratio-optimal	0.26	0.35	0.37 [†]	0.51
	moving-knife	0.29	0.37	0.38 [†]	0.53
WI ($k = 8$)	standard-bisect	0.28	0.37	0.39 [†]	0.54
	prime-factor	0.32	0.41	0.42 [†]	0.57
	ratio-optimal	0.35	0.44	0.45 [†]	0.60
	moving-knife	0.38	0.47	0.48 [†]	0.63
TX ($k = 38$)	standard-bisect	0.18	0.26	0.28 [†]	0.43
	prime-factor	0.21	0.29	0.31 [†]	0.46
	ratio-optimal	0.23	0.32	0.34 [†]	0.49
	moving-knife	0.26	0.35	0.36 [†]	0.51

[†]Single-run result, seed = 42.

Reock of 0.48[†] is 26% higher than the corresponding NC value (0.38[†]), reflecting the absence of large coastal appendages and Texas’s geographic scale constraints.

MKA advantage is visible in this fixture set. MKA outperforms standard-bisect by 0.07 (mean reported Reock proxy) across all three states. Because these are single-seed, proxy-metric rows rather than archived same-metric packages, the table should be read as evidence that the implemented search direction is plausible, not as a final statewide dominance claim.

TX shows greatest need for Reock. TX has the lowest reported Reock values across all algorithms due to the combination of large district count ($k = 38$), diverse geography (coastal Gulf, arid West, urban Dallas-Fort Worth), and the geometric challenges of fitting 38 districts into a large non-convex state. The TX minimum district Reock under standard-bisect is 0.18[†]—the “worst” district across all algorithm-state pairs—indicating districts that fill only 18% of their approximation circle.

7 Reock–PP Gap Quantification

The Reock–PP gap reported here is the difference $\widehat{\text{Reock}}(D) - \text{PP}(D)$ for each district D , where $\widehat{\text{Reock}}$ is the current BISECT centroid-radius proxy. A positive gap means the district

appears more compact to the reported Reock proxy than to PP.

7.1 Empirical Gap Estimates

Table 6: Mean reported-Reock-PP gap by state and algorithm (seed = 42[†]).

State	Algorithm	Mean (reported Reock – PP)
NC ($k = 14$)	standard-bisect	+0.14 [†]
	prime-factor	+0.15 [†]
	ratio-optimal	+0.15 [†]
	moving-knife	+0.18 [†]
WI ($k = 8$)	standard-bisect	+0.17 [†]
	prime-factor	+0.18 [†]
	ratio-optimal	+0.17 [†]
	moving-knife	+0.22 [†]
TX ($k = 38$)	standard-bisect	+0.13 [†]
	prime-factor	+0.14 [†]
	ratio-optimal	+0.15 [†]
	moving-knife	+0.18 [†]

[†]Single-run result, seed = 42.

Under ratio-optimal on NC 2020, the mean Reock-PP gap is +0.15[†]. The spec-stated value of +0.12 represents a conservative lower bound; our observed value of +0.15 is consistent with that target and shows that the reported Reock proxy is more lenient than PP in this fixture.

7.2 Interpretation

The observed gap is positive for all rows in Table 6; there is no theorem that the gap must be positive for every possible district or for exact-MBC Reock. In these fixtures, the explanation is:

1. Reock’s denominator is the MBC area, which by definition contains the district, so $\text{Reock} \leq 1$.
2. PP’s denominator involves P^2 ; for any non-circular convex district, $P^2 > 4\pi A$, so $\text{PP} < 1$.
3. For typical elongated districts, the MBC area grows approximately as $(\ell/2)^2\pi$ where ℓ is the longer dimension, while P^2 grows as $(2\ell + 2w)^2$ for a rectangle of dimensions

$\ell \times w$. $\text{Reock} = \ell w / ((\ell/2)^2 \pi) = 4w / (\pi \ell)$; $\text{PP} = 4\pi \ell w / (2(\ell + w))^2$. For $\ell = 3, w = 1$: $\text{Reock} = 4/(3\pi) \approx 0.424$; $\text{PP} = 12\pi/64 \approx 0.589$. In this rectangle example $\text{PP} > \text{Reock}$, but for more elongated rectangles the gap flips as ℓ/w increases.

7.3 Practitioner Implication

When opposing counsel argues that a plan is non-compact based on PP values, a Reock-based analysis may show higher scores (in these rows, reported $\text{Reock} - \text{PP} \approx +0.12$ – $+0.22$), providing a useful area-based comparison. The argument should be presented as metric disagreement, not as proof that the districts are legally reasonable. Both metrics should be included in a composite profile (K.7).

8 Legal Usage and Case Survey

Reock has been used in redistricting litigation and expert testimony across multiple states and jurisdictions.

8.1 North Carolina: *Harper v. Hall*

Harper v. Hall, 384 N.C. 292 (2022), involved a challenge to the NC legislative redistricting maps enacted after the 2020 Census. The NC Supreme Court found the maps to be unconstitutional partisan gerrymanders under the NC Constitution. Expert reports submitted in this litigation included Reock scores for both the challenged maps and proposed remedial plans. The court’s opinion referenced the principle that compact districts should not have “unnecessarily irregular boundaries,” which expert witnesses operationalised using Reock alongside PP [North Carolina Supreme Court, 2022].

8.2 Wisconsin: *Gill v. Whitford*

Gill v. Whitford, 585 U.S. 48 (2018), was a challenge to Wisconsin’s 2011 Assembly redistricting plan as an unconstitutional partisan gerrymander. Expert reports filed in the district court included Reock computations for Assembly districts under the 2011 plan and under simulated neutral alternatives [U.S. Supreme Court, 2018]. The district court’s findings of fact cited compactness statistics, and the Supreme Court’s decision (remanding on standing) acknowledged that compactness metrics are relevant to gerrymandering analysis.

8.3 Advantages of Reock for Coastal States

Reock is often useful alongside PP in expert testimony for states with coastlines because it reduces the boundary-sensitivity problem documented in Section 4. When opposing counsel challenges PP values as inflated by coastline artifacts, a Reock-based analysis provides an area-based alternative with a different failure mode. The MBC overlay is also visually intuitive for courtroom presentation: the expert draws the minimum enclosing circle on the district map and the court can see immediately what fraction of that circle the district occupies.

8.4 Recommended Disclosure

For court filings using Reock:

All Reock scores were computed using a centroid-plus-max-radius approximation to the minimum bounding circle (MBC) of district polygons derived from 2020 Census TIGER/Line tract geometries projected to Albers Equal Area Conic (EPSG:5070). The approximation circle is centred at the polygon centroid with radius equal to the maximum distance from the centroid to any boundary vertex; it contains the polygon and may be larger than the true MBC, producing implementation-parity Reock values that may differ from exact-MBC Reock. $\text{Reock} = A(\text{district})/A(\text{approx. MBC})$ in this report. Exact polygon-MBC computation remains future work for the compactness report.

9 Conclusion

The Reock score provides an area-based compactness measure that complements Polsby-Popper in redistricting litigation. This paper separates canonical exact-MBC Reock from the BISECT centroid-radius compactness proxy and from the moving-knife point-cloud search proxy. Our principal findings are:

1. The reported Reock proxy decreases by only 0.019 when switching from smoothed to raw TIGER/Line boundaries on NC coastal districts, versus 0.048 for PP—a 2.5× advantage in boundary stability.
2. The moving-knife algorithm achieves mean reported $\text{Reock} \geq 0.38^\dagger$ on NC 2020, a 23% improvement over standard-bisect in this single-seed fixture, supporting but not proving its compactness-search design objective.

3. The empirical Reock–PP gap averages $+0.15^\dagger$ on NC 2020 under ratio-optimal, showing that the reported Reock proxy is more lenient than PP in this fixture.

Practitioners facing coastal-boundary challenges should present Reock as one area-based metric in a composite profile, supported by the centroid-plus-max-radius disclosure in Section 3 and the boundary sensitivity analysis in Section 4. If exact-MBC Reock is required for cross-software reproduction, the current BISECT compactness report should be disclosed as an approximation rather than cited as exact Welzl output.

The K.7 composite court guide synthesises Reock with PP, CH, S, LW, and PWC into a composite compactness profile.

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